



Modification of a generalized three-dimensional Hoek–Brown strength criterion

Qi Zhang^{a,b,c,d}, Hehua Zhu^{a,b}, Lianyang Zhang^{d,*}

^a Department of Geotechnical Engineering, Tongji University, Shanghai, China

^b Key Laboratory of Geotechnical and Underground Engineering, Tongji University, Ministry of Education, Shanghai, China

^c State Key Laboratory for GeoMechanics and Deep Underground Engineering, China University of Mining & Technology, Xuzhou, China

^d Department of Civil Engineering and Engineering Mechanics, University of Arizona, Tucson, AZ, USA

ARTICLE INFO

Article history:

Received 3 December 2011

Received in revised form

18 October 2012

Accepted 19 December 2012

Available online 19 January 2013

Keywords:

Hoek–Brown strength criterion

3D strength criterion

Failure surface

Lode dependence

Smoothness

Convexity

ABSTRACT

To take account of the influence of the intermediate principle stress, Zhang and Zhu [1,2] proposed a three-dimensional (3D) version of the generalized Hoek–Brown strength criterion. The generalized Zhang–Zhu strength criterion is a true 3D version of the Hoek–Brown criterion, not only inheriting the advantages of the Hoek–Brown strength criterion, but predicts the same strength as the two-dimensional (2D) Hoek–Brown strength criterion at both triaxial compression and extension states. However, the failure surface of the generalized 3D Zhang–Zhu strength criterion is not smooth at either the triaxial compression or extension state and concave at the triaxial extension state, which may have problems with some stress paths and cause inconvenience for numerical applications. In this paper, the reason for the non-smoothness and non-convexity of the generalized 3D Zhang–Zhu strength criterion was first discussed by studying its Lode dependence. Then the criterion was modified by utilizing three different Lode dependences with characteristics of both smoothness and convexity to replace its Lode dependence. Finally the smoothness, convexity and prediction accuracy of the modified criteria were evaluated by applying them to analyze both intact rocks and jointed rock masses. The modified criteria not only keep the advantages of the generalized 3D Zhang–Zhu strength criterion, but solve the non-smoothness and non-convexity problem with no loss of accuracy for strength prediction.

© 2012 Elsevier Ltd. All rights reserved.

1. Introduction

Researchers have developed different strength criteria for rock [1–9]. Of these different strength criteria, the Hoek–Brown strength criterion has been used most widely. The Hoek–Brown strength criterion was developed specifically for rock materials and rock masses, and its input parameters can be determined from routine unconfined compression tests, mineralogical examination, and discontinuity characterization [4]. It has been applied successfully to a wide range of intact and fractured rock types during the past three decades.

The Hoek–Brown failure criterion, however, does not take account of the influence of the intermediate principal stress, although much evidence has been accumulating to indicate that the intermediate principal stress does influence the rock strength in many instances. Therefore, researchers have developed 3D versions of the Hoek–Brown strength criterion. The 3D versions of the Hoek–Brown strength criterion have advantages over other existing 3D non-Hoek–Brown type strength criteria in that they use the same input parameters as the widely used Hoek–Brown strength criterion [1,2,17,20]. Of the 3D Hoek–Brown strength criteria, only the Zhang–Zhu criterion predicts the same strength as the original Hoek–Brown strength criterion at both triaxial compression and extension states; but due to the non-smoothness of the failure surface at both triaxial compression and extension states and the non-convexity of the failure surface at the triaxial extension state, the Zhang–Zhu criterion may have problems with some stress paths and cause inconvenience in numerical applications

(see next section about more detailed review of the Hoek–Brown strength criterion and different 3D Hoek–Brown strength criteria).

In this paper, the generalized 3D Zhang–Zhu criterion was modified by applying three Lode dependences to address the problem of non-smoothness and non-convexity. The modified criteria were then used to analyze both intact rocks and jointed rock masses.

* Corresponding author. Tel.: +1 520 6260532; fax: +1 520 6212550.

E-mail address: lyzhang@email.arizona.edu (L. Zhang).

The modified criteria not only keep the advantages of the generalized Zhang–Zhu criterion, but solve the non-smoothness and non-convexity problem without loss of accuracy for strength prediction.

2. Hoek–Brown strength criterion and its 3D versions

This section describes the Hoek–Brown strength criterion and its 3D versions and performs a brief evaluation of these strength criteria, which provides the background information for modification of the generalized Zhang–Zhu criterion in next section.

2.1. Original Hoek–Brown strength criterion

The Hoek–Brown strength criterion was originally developed for intact rock and then extended to rock masses. The process used by Hoek and Brown in deriving their strength criterion for intact rock was one of pure trial and error. Apart from the conceptual starting point provided by the Griffith theory, there is no fundamental relationship between the empirical constants included in the criterion and any physical characteristics of the rock. The justification for choosing this particular criterion over the numerous alternatives lies in the adequacy of its predictions of the observed rock fracture behavior, and the convenience of its application to a range of typical engineering problems [10].

For intact rock, the Hoek–Brown strength criterion may be expressed in the following form

$$\sigma_1 = \sigma_3 + \sigma_c \left(m_i \frac{\sigma_3}{\sigma_c} + 1 \right)^{0.5} \quad (1)$$

where σ_c is the unconfined compression strength of the intact rock; σ_1 and σ_3 are respectively the major and minor effective principal stresses; and m_i is a material constant for the intact rock, which depends upon the rock type (texture and mineralogy) as tabulated in Table 1.

Table 1
Values of parameter m_i for different rocks [1, 2, 33, 34].

Rock type	Class	Group	Texture			
			Coarse	Medium	Fine	Very fine
Sedimentary	Clastic		Conglomerate (21 ± 3) ^a Breccia (19 ± 5)	Sandstone 17 ± 4	Siltstone 7 ± 2 Greywacke (18 ± 3)	Claystone 4 ± 2 Shale (6 ± 2) Marl (7 ± 2) Dolomite (9 ± 3)
	Non-clastic	Carbonate	Crystalline limestone (12 ± 3)	Sparitic limestone (10 ± 2)	Micritic limestone (9 ± 2)	
		Evaporite		Gypsum 8 ± 2	Anhydrite 12 ± 2	
Metamorphic	Non-foliated		Marble 9 ± 3	Hornfels (19 ± 4) Metasandstone (19 ± 3)	Quartzite 20 ± 3	Chalk 7 ± 2
				Amphibolite 26 ± 6	Gneiss 28 ± 5	
	Slightly foliated		Migmatite (29 ± 3)	Schist 12 ± 3	Phyllite (7 ± 3)	Slate 7 ± 4
	Foliated ^b			Diorite 25 ± 5		
Igneous	Plutonic	Light	Granite 32 ± 3 Granodiorite (29 ± 3)	Dolerite (16 ± 5)		
		Dark	Gabbro 27 ± 3 Norite 20 ± 5			
	Hypabyssal		Porphyrie (20 ± 5)		Diabase (15 ± 5)	Peridotite (25 ± 5)
	Volcanic	Lava		Rhyolite (25 ± 5) Andesite 25 ± 5	Dacite (25 ± 3) Basalt (25 ± 5)	Obsidian (19 ± 3)
				Breccia (19 ± 5)	Tuff (13 ± 5)	
		Pyroclastic	Agglomerate (19 ± 3)			

^a Values in parenthesis are estimates.

^b These values are for intact rock specimen tests normal to bedding or foliation. The value of m_i will be significantly different if failure occurs along a weakness plane.

For jointed rock masses, the general form of the Hoek–Brown criterion, which incorporates both the original and the modified forms, is given by

$$\sigma_1 = \sigma_3 + \sigma_c \left(m_b \frac{\sigma_3}{\sigma_c} + s \right)^a \quad (2)$$

where m_b is the material constant for the rock masses; and s and a are constants that depend on the characteristics of the rock masses. Hoek and Brown [11] proposed a set of relations between the parameters m_b , s and a and the 1976 version of Bieniawski's Rock Mass Rating (RMR), assuming completely dry conditions and a very favorable (according to the RMR rating system) discontinuity orientation:

(i) Disturbed rock masses

$$m_b = \exp\left(\frac{\text{RMR}-100}{14}\right) m_i \quad (3a)$$

$$s = \exp\left(\frac{\text{RMR}-100}{6}\right) \quad (3b)$$

$$a = 0.5 \quad (3c)$$

(ii) Undisturbed or interlocking rock masses

$$m_b = \exp\left(\frac{\text{RMR}-100}{28}\right) m_i \quad (4a)$$

$$s = \exp\left(\frac{\text{RMR}-100}{9}\right) \quad (4b)$$

$$a = 0.5 \quad (4c)$$

Eqs. (3) and (4) are acceptable for rock masses with RMR values greater than about 25, but do not work for very poor rock masses since the minimum value which RMR can assume is 18 for the 1976 RMR system and 23 for the 1989 RMR system. In order to overcome this limitation, Hoek [12] and Hoek et al. [13] introduced the Geological Strength Index (GSI). The relationships between m_b , s and a and GSI are as follows:

(i) For $\text{GSI} > 25$, i.e. rock masses of good to reasonable quality

$$m_b = \exp\left(\frac{\text{GSI}-100}{28}\right) m_i \quad (5a)$$

$$s = \exp\left(\frac{\text{GSI}-100}{9}\right) \quad (5b)$$

$$a = 0.5 \quad (5c)$$

(ii) For $\text{GSI} < 25$, i.e. rock masses of very poor quality

$$m_b = \exp\left(\frac{\text{GSI}-100}{28}\right) m_i \quad (6a)$$

$$s = 0 \quad (6b)$$

$$a = 0.65 - \frac{\text{GSI}}{200} \quad (6c)$$

Hoek et al. [14] proposed new relationships between m_b , s and a and GSI by introducing a new parameter D , which is a factor that depends on the degree of disturbance due to blast damage and stress relaxation. The values of D range from 0 for undisturbed in situ rock masses to 1 for very disturbed rock masses. The guidelines for selecting D can be found in Hoek [15].

$$m_b = \exp\left(\frac{\text{GSI}-100}{28-14D}\right) m_i \quad (7a)$$

$$s = \exp\left(\frac{\text{GSI}-100}{9-3D}\right) \quad (7b)$$

$$a = 0.5 + \frac{1}{6} [\exp(-\text{GSI}/15) - \exp(-20/3)] \quad (7c)$$

It is noted that the distinction between disturbed and undisturbed rock masses is dropped in evaluating the parameters m_b , s and a from GSI. This is based on the fact that disturbance is generally induced by engineering activities and should be allowed by downgrading the values of GSI.

2.2. Three-dimensional Hoek–Brown strength criteria

A major limitation for the Hoek–Brown strength criterion is that it does not consider the effect of the intermediate principal stress, although it has been found that the intermediate principal stress influences the rock strength in many instances [6–9,16–19]. Therefore, researchers have developed 3D versions of the Hoek–Brown strength criterion. The 3D versions of the Hoek–Brown strength criterion have advantages over other existing 3D non-Hoek–Brown type criteria in that they use the same input parameters as the widely used Hoek–Brown criterion. The following briefly describes several 3D Hoek–Brown strength criteria.

2.2.1. Pan–Hudson criterion

Pan and Hudson [17] proposed a 3D version of the original Hoek–Brown strength criterion for rock mass (Eq. (2) with $a=0.5$), which is expressed as

$$\frac{9}{2\sigma_c} \tau_{oct}^2 + \frac{3}{2\sqrt{2}} m_b \tau_{oct} - m_b \frac{I_1}{3} = s\sigma_c \quad (8)$$

where τ_{oct} and I_1 are, respectively, the octahedral shear stress and the first stress invariant defined by

$$\tau_{oct} = \frac{1}{3} \sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2} \quad (9)$$

$$I_1 = \sigma_1 + \sigma_2 + \sigma_3 \quad (10)$$

in which σ_1 , σ_2 and σ_3 are, respectively, the major, intermediate and minor effective principal stresses.

2.2.2. Generalized Priest criterion

Priest [20] developed a 3D version of the generalized Hoek–Brown strength criterion (Eq. (2)) by combining it with the Drucker–Prager criterion, which is expressed by

$$J_2^{1/2} = AJ_1 + B \quad (11)$$

where A and B are empirical parameters; $J_1 = I_1/3$ is the mean effective stress; and J_2 is the second deviatoric stress invariant defined by

$$J_2 = \frac{1}{6} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2] \quad (12)$$

Since the 3D stress state satisfying the generalized Hoek–Brown strength criterion (Eq. (2)) is $(\sigma_1, \sigma_3, \sigma_3)$, the strategy is to determine parameters A and B for the Drucker–Prager failure surface that intersects the Hoek–Brown failure point $(\sigma_1, \sigma_3, \sigma_3)$. With parameters A and B determined, the strength σ_{zf} for an element of material subjected to a general state of 3D stresses $\sigma_x \neq \sigma_y \neq \sigma_{zf}$ can then be predicted. This process is equivalent to the well known process identifying the Drucker–Prager parameters that give the circumscribed fit for the Coulomb strength criterion. Priest [20] originally suggested a numerical procedure for determining parameters A and B and strength σ_{zf} .

Melkounian et al. [21] later developed an explicit solution for determining strength σ_{zf} at intermediate and minor principal stresses σ_x and σ_y , which is summarized below

$$\sigma_{zf} = 3\sigma_p + \sigma_c \left[\frac{m_b \sigma_p}{\sigma_c} + s \right]^a - (\sigma_x + \sigma_y) \quad (13)$$

in which

$$\sigma_p = \frac{\sigma_x + \sigma_y}{2} + \frac{-E + \sqrt{E^2 - F(\sigma_x - \sigma_y)^2}}{2F} \quad (14)$$

where $E = 2C^a \sigma_c$, $F = 3 + 2aC^{a-1} m_b$, $C = s + \frac{m_b(\sigma_x + \sigma_y)}{2\sigma_c}$.

2.2.3. Zhang–Zhu criterion

Zhang and Zhu [1] proposed a 3D version of the original Hoek–Brown strength criterion for rock mass (Eq. (2) with $a=0.5$) by combining the general Mogi [16] criterion and the Hoek–Brown strength criterion, which is expressed as

$$\frac{9}{2\sigma_c} \tau_{oct}^2 + \frac{3}{2\sqrt{2}} m_b \tau_{oct} - m_b \sigma_{m,2} = s\sigma_c \quad (15)$$

where τ_{oct} is the octahedral shear stress as defined by Eq. (9), and $\sigma_{m,2}$ is the mean stress defined by

$$\sigma_{m,2} = \frac{\sigma_1 + \sigma_3}{2} \quad (16)$$

2.2.4. Generalized Zhang–Zhu criterion

Zhang [2] proposed a 3D version of the generalized Hoek–Brown strength criterion (Eq. (2)) by modifying Eq. (15), which is expressed as

$$\frac{1}{\sigma_c^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{oct} \right)^{1/a} + \frac{m_b}{2} \left(\frac{3}{\sqrt{2}} \tau_{oct} \right) - m_b \sigma_{m,2} = s \sigma_c \quad (17)$$

where τ_{oct} and $\sigma_{m,2}$ are, respectively, the octahedral shear stress and the mean stress defined by Eqs. (9) and (16).

2.3. Brief evaluation of 3D versions of Hoek–Brown strength criterion

The (generalized) 3D Zhang–Zhu criterion predicts the same strength as the two-dimensional (2D) Hoek–Brown strength criterion at both triaxial compression ($\sigma_2 = \sigma_3$) and extension ($\sigma_1 = \sigma_2$) stress states; but the 3D Pan–Hudson criterion does not predict the same strength as the Hoek–Brown criterion at either triaxial compression or extension stress state. Since the 3D Priest criterion is derived by identifying the Drucker–Prager parameters that give the circumscribed fit for the Hoek–Brown criterion, it predicts the same strength as the Hoek–Brown strength criterion at triaxial compression stress state but not at triaxial extension stress state [1,2]. So the (generalized) 3D Zhang–Zhu criterion tends to provide better strength predictions, especially at the triaxial compression and extension stress states. However, it is noted that the (generalized) Zhang–Zhu criterion envelope is not smooth at either the triaxial compression or extension stress state and concave at the triaxial extension stress state, which may lead to problems with some stress paths and cause inconvenience in numerical applications [1,2].

3. Presentation of strength criterion in terms of I_1 , J_2 and θ_σ

A strength criterion can also be presented in terms of I_1 , J_2 and θ_σ , where θ_σ is the Lode angle, which is determined by

$$\theta_\sigma = \tan^{-1} \left[\frac{\sigma_1 + \sigma_3 - 2\sigma_2}{\sqrt{3}(\sigma_1 - \sigma_3)} \right] \quad (18)$$

The Lode angle θ_σ ranges from $\pi/6$ to $-\pi/6$. When $\sigma_1 > \sigma_2 = \sigma_3$, the stress condition is triaxial compression and the Lode angle θ_σ is equal to $\pi/6$. When $\sigma_1 = \sigma_2 > \sigma_3$, the strength condition is triaxial extension and the Lode angle θ_σ is equal to $-\pi/6$.

Using the Lode angle θ_σ , the major, intermediate and minor principal stresses, σ_1 , σ_2 and σ_3 , can be derived as

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{bmatrix} = -\frac{2}{\sqrt{3}} \sqrt{J_2} \begin{bmatrix} \sin(\theta_\sigma - \frac{2}{3}\pi) \\ \sin(\theta_\sigma) \\ \sin(\theta_\sigma + \frac{2}{3}\pi) \end{bmatrix} + \begin{bmatrix} I_1/3 \\ I_1/3 \\ I_1/3 \end{bmatrix} \quad (19)$$

where I_1 and J_2 are respectively the first stress invariant and the second deviatoric stress invariant as defined earlier.

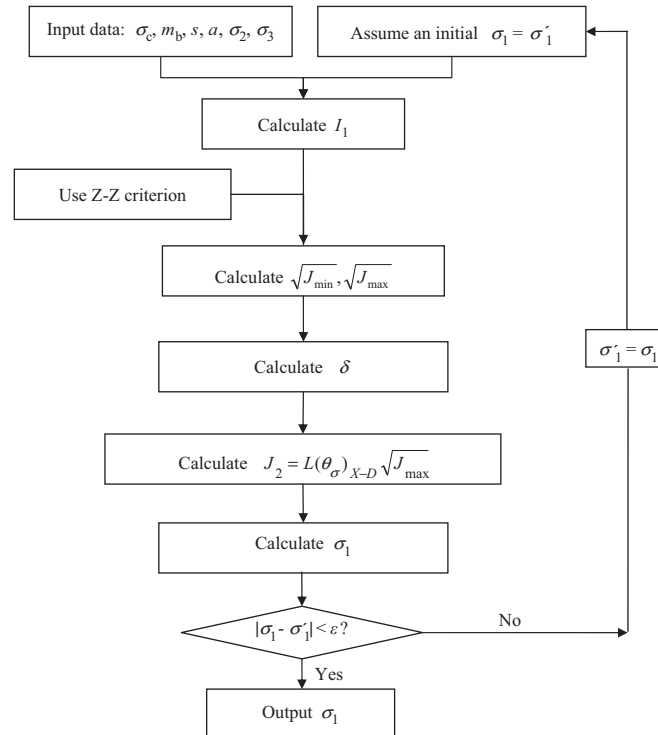


Fig. 1. Flowchart for determining strength value at $a \neq 0.5$ with modified criteria. ε is the iterative convergence limit, which is set as 0.001 MPa in this paper.

With the principal stresses represented by Eq. (19), the strength criteria described in the previous section can be presented in terms of I_1, J_2 and θ_σ as follows.

3.1. Generalized Hoek–Brown criterion

Combining Eqs. (2) and (19), the generalized Hoek–Brown criterion in terms of I_1, J_2, θ_σ can be expressed as

$$F = \frac{1}{\sigma_c^{(1/a-1)}} \left(2 \cos \theta_\sigma \sqrt{J_2} \right)^{1/a} + \left(\cos \theta_\sigma - \frac{\sin \theta_\sigma}{\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s \sigma_c = 0 \quad (20)$$

When $a=0.5$, Eq. (20) reduces to the original Hoek–Brown strength criterion in terms of I_1, J_2, θ_σ

$$F = \frac{4 \cos^2 \theta_\sigma}{\sigma_c} J_2 + \left(\cos \theta_\sigma - \frac{\sin \theta_\sigma}{\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s \sigma_c = 0 \quad (21)$$

3.2. 3D Pan–Hudson criterion

Combining Eqs. (8) and (19), the Pan–Hudson criterion in terms of I_1, J_2, θ_σ can be expressed as

$$F = \frac{3}{\sigma_c} J_2 + \frac{\sqrt{3}}{2} m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s \sigma_c = 0 \quad (22)$$

3.3. Generalized 3D Zhang–Zhu criterion

Combining Eqs. (17) and (19), the generalized 3D Zhang–Zhu criterion in terms of I_1, J_2, θ_σ can be expressed as

$$F = \frac{1}{\sigma_c^{(1/a-1)}} \left(\sqrt{3} J_2 \right)^{1/a} + \left(\frac{\sqrt{3}}{2} - \frac{\sin \theta_\sigma}{\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s \sigma_c = 0 \quad (23)$$

When $a=0.5$, Eq. (23) will be the Zhang–Zhu criterion in terms of I_1, J_2, θ_σ :

$$F = \frac{3}{\sigma_c} J_2 + \left(\frac{\sqrt{3}}{2} - \frac{\sin \theta_\sigma}{\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s \sigma_c = 0 \quad (24)$$

Comparing Eqs. (20) and (23), it is interesting to note that Eq. (23) is a special case of Eq. (20) when $\cos \theta_\sigma$ is equal to $\sqrt{3}/2$. At both triaxial compression ($\theta_\sigma = \pi/6$) and extension ($\theta_\sigma = -\pi/6$) states, $\cos \theta_\sigma = \sqrt{3}/2$ and thus Eqs. (20) and (23) predict the same strength. This is another way to explain why the (generalized) 3D Zhang–Zhu criterion predicts the same strength as the Hoek–Brown criterion at both triaxial compression and extension states.

If we compare Eqs. (22) and (24), it can be seen that Eq. (22) is a special case of Eq. (24) when $\sin \theta_\sigma$ equals 0. So the Pan–Hudson criterion predicts the same strength as the Zhang–Zhu criterion at Lode angle θ_σ equal to 0; but the Pan–Hudson criterion under-predicts the strength at Lode angle equal θ_σ to $\pi/6$ (triaxial compression stress state) and over-predicts the strength at Lode angle θ_σ equal to $-\pi/6$ (triaxial extension stress state).

4. Modification of the generalized 3D Hoek–Brown strength criterion

Form the study in the previous section, it can be seen that the Lode dependence is a key in controlling the shape of the failure surface of a strength criterion. The following discusses the Lode dependence in more detail and then modifies the generalized 3D Zhang–Zhu criterion to address its problem of non-smoothness and non-convexity.

4.1. Requirements for Lode dependence for a strength criterion

The Lode dependence generally needs to satisfy and obey the requirements for aspect ratio, smoothness, and convexity [22] as described below.

4.1.1. Requirement for aspect ratio

The requirement on aspect ratio is expressed as

$$L(-\pi/6) = \delta, \quad L(\pi/6) = 1 \quad (25)$$

where $L(\theta_\sigma)$ is the Lode dependence and varies between 0.5 and 1; and δ is the aspect ratio which is the ratio of the Lode dependence at $\theta_\sigma = -\pi/6$ to that at $\theta_\sigma = \pi/6$. The value of δ controls the general shape of the deviatoric sections of the failure surface.

4.1.2. Requirement for smoothness

The failure surface is smooth at $\theta_\sigma = \pm \pi/6$ when

$$L'(\pm \pi/6) = 0 \quad (26)$$

provided that $L(\theta_\sigma)$ is differentiable.

4.1.3. Requirement on convexity

The failure surface of a strength criterion is convex in deviatoric planes when [23,24]

$$\frac{d^2L(\theta_\sigma)}{d\theta_\sigma^2} < L(\theta_\sigma) + \frac{2}{L(\theta_\sigma)} \left(L'(\theta_\sigma) \right)^2 \tag{27}$$

The convexity requirement is useful to satisfy the stability postulate of Drucker [25].

4.2. Lode dependence for the generalized 3D Zhang–Zhu criterion

With the generalized 3D Zhang–Zhu strength criterion as expressed by Eq. (23), J_2 can be determined at different Lode angles and the Lode dependence ratio can then be determined by

$$\delta = \sqrt{J_{\min}/J_{\max}} = \sqrt{J_2(-\pi/6)/J_2(\pi/6)} \tag{28}$$

When $a \neq 0.5$, an explicit expression of $\sqrt{J_2}$ with different Lode angles cannot be derived at present and thus a numerical solution strategy will be presented. For a special case $a=0.5$, $\sqrt{J_2}$ can be explicitly derived as

$$\sqrt{J_2} = \frac{\sqrt{\left(4\sin^2\theta_\sigma - 12\sin\theta_\sigma + 9\right)m_b^2\sigma_c^2 + 48m_b\sigma_cI_1 + 144s\sigma_c^2 - (3 - 2\sin\theta_\sigma)m_b\sigma_c}}{12\sqrt{3}} \tag{29}$$

When $\theta_\sigma = \pi/6$, $\sin\theta_\sigma = 0.5$ and $J_2 = J_{\max}$ which can be determined by

$$\sqrt{J_{\max}} = \frac{1}{6\sqrt{3}} \left(\sqrt{m_b^2\sigma_c^2 + 12m_b\sigma_cI_1 + 36s\sigma_c^2} - m_b\sigma_c \right) \tag{30}$$

Similarly when $\theta_\sigma = -\pi/6$, $\sin\theta_\sigma = -0.5$ and $J_2 = J_{\min}$ which can be determined by

$$\sqrt{J_{\min}} = \frac{1}{3\sqrt{3}} \left(\sqrt{m_b^2\sigma_c^2 + 3m_b\sigma_cI_1 + 9s\sigma_c^2} - m_b\sigma_c \right) \tag{31}$$

With the derived Eqs. (30) and (31), the aspect ratio can be derived as

$$\delta = \sqrt{\frac{J_{\min}}{J_{\max}}} = \frac{2\sqrt{m_b^2\sigma_c^2 + 3m_b\sigma_cI_1 + 9s\sigma_c^2} - 2m_b\sigma_c}{\sqrt{m_b^2\sigma_c^2 + 12m_b\sigma_cI_1 + 36s\sigma_c^2} - m_b\sigma_c} \tag{32}$$

Table 2
Input data for the three cases for validating smoothness and convexity.

Fig.	GSI	m_i	D	m_b^*	s^*	a^*	I_1 (MPa)	Type
2	100	15	0	15	1	0.5	300	Intact rock
3	75	15	0	6.1423	0.0622	0.5009	300	Rock mass
4	40	15	0	1.7598	0.0013	0.5114	300	Rock mass

* Calculated from Eq. (7).

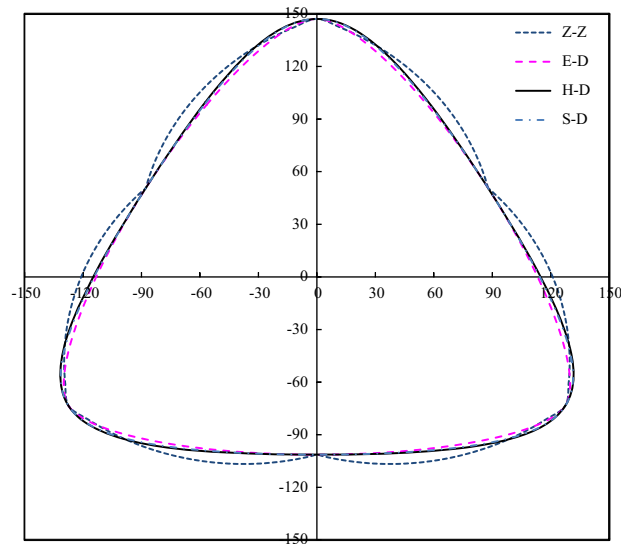


Fig. 2. Failure surface plots of generalized Zhang–Zhu (Z-Z) criterion and three modified criteria with elliptical dependence (E-D), hyperbolic dependence (H-D), and spatial mobilized plane dependence (S-D) at $I_1=300$ MPa for an intact rock (GSI=100) with input data listed in Table 2.

Similarly, the Lode dependence of the generalized 3D Zhang–Zhu strength criterion at $a=0.5$ can be derived as

$$L(\theta_\sigma) = \frac{\sqrt{J_2}}{\sqrt{J_{\max}}} = \frac{\sqrt{(4\sin^2\theta_\sigma - 12\sin\theta_\sigma + 9)m_b^2\sigma_c^2 + 48m_b\sigma_c I_1 + 144s\sigma_c^2 - (3 - 2\sin\theta_\sigma)m_b\sigma_c}}{2(\sqrt{m_b^2\sigma_c^2 + 12m_b\sigma_c I_1 + 36s\sigma_c^2} - m_b\sigma_c)} \quad (33)$$

and the differentiability of the Lode dependence can be expressed as

$$L'(\theta_\sigma) = \frac{\left((8\sin\theta_\sigma\cos\theta_\sigma - 12\cos\theta_\sigma)m_b^2\sigma_c^2 / \left(2\sqrt{(4\sin^2\theta_\sigma - 12\sin\theta_\sigma + 9)m_b^2\sigma_c^2 + 48m_b\sigma_c I_1 + 144s\sigma_c^2} \right) \right) + 2\cos\theta_\sigma m_b\sigma_c}{2(\sqrt{m_b^2\sigma_c^2 + 12m_b\sigma_c I_1 + 36s\sigma_c^2} - m_b\sigma_c)} \quad (34)$$

with the derived Eq. (34), $L'(\pm\pi/6) \neq 0$ and thus the smoothness requirement is not met at either the triaxial compression or extension stress state. Further evaluation indicates that the convexity requirement is also not met at $\theta_\sigma = -\pi/6$. The non-convexity is a common problem of the Mogi [16] criterion and the criteria based on the concept of Mogi [26,27].

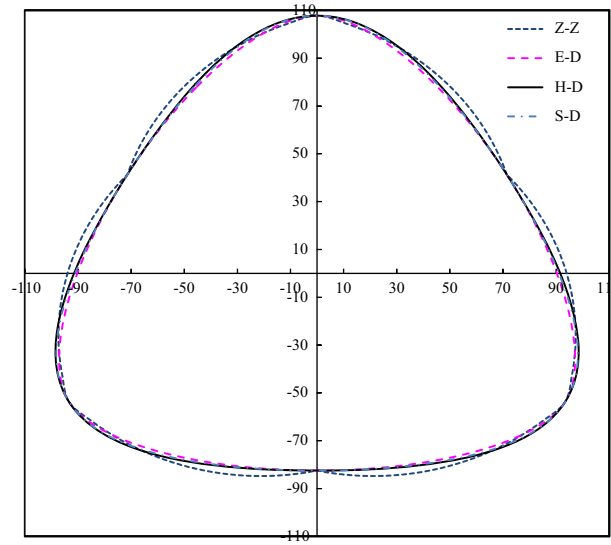


Fig. 3. Failure surface plots of generalized Zhang–Zhu (Z-Z) criterion and three modified criteria with elliptical dependence (E-D), hyperbolic dependence (H-D), and spatial mobilized plane dependence (S-D) at $I_1 = 300$ MPa for a rock mass (GSI = 75) with input data listed in Table 2.

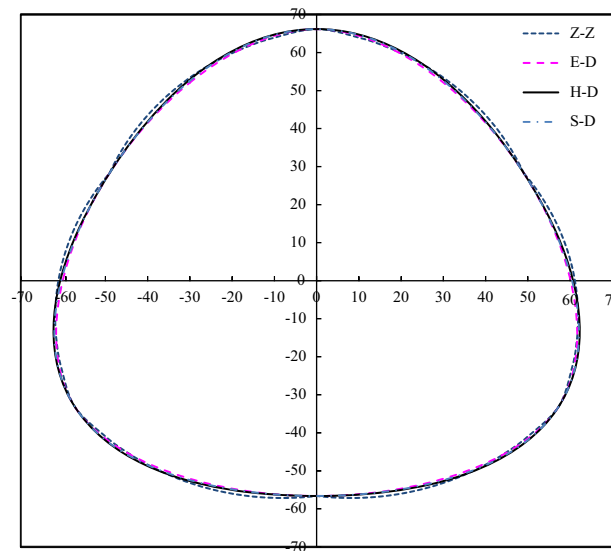


Fig. 4. Failure surface plots of generalized Zhang–Zhu (Z-Z) criterion and three modified criteria with elliptical dependence (E-D), hyperbolic dependence (H-D), and spatial mobilized plane dependence (S-D) at $I_1 = 300$ MPa for a rock mass (GSI = 40) with input data listed in Table 2.

4.3. Modification of the generalized 3D Zhang–Zhu criterion

To address the non-smoothness and non-convexity problem for the generalized 3D Zhang–Zhu criterion, we use other Lode dependences with smoothness and convexity characteristics to replace the Lode dependence of the generalized 3D Zhang–Zhu criterion. Researchers have proposed different Lode dependences. The following briefly describes three Lode dependences that possess both smoothness and convexity characteristics and are used in this paper.

The dependence proposed by William and Warnke [28] uses an elliptical approximation of the deviatoric section (E-D) and is expressed as

$$L(\theta_{\sigma})_{E-D} = \frac{2\left(1-\delta^2\right)\cos(\pi/6+\theta_{\sigma})+(2\delta-1)\sqrt{4\left(1-\delta^2\right)\cos^2(\pi/6+\theta_{\sigma})+\delta(5\delta-4)}}{4\left(1-\delta^2\right)\cos^2(\pi/6+\theta_{\sigma})+(2\delta-1)^2} \tag{35}$$

Yu [29] proposed a hyperbolic dependence (H-D) which is expressed by

$$L(\theta_{\sigma})_{H-D} = \frac{2\delta\left(1-\delta^2\right)\cos(\pi/6-\theta_{\sigma})+\delta(\delta-2)\sqrt{4\left(\delta^2-1\right)\cos^2(\pi/6-\theta_{\sigma})+(5-4\delta)}}{4\left(1-\delta^2\right)\cos^2(\pi/6-\theta_{\sigma})-(\delta-2)^2} \tag{36}$$

The dependence proposed by Matsuoka and Nakai [30] is based on the spatial mobilized plane (S-D) and is expressed as

$$L(\theta_{\sigma})_{S-D} = \frac{\sqrt{3}\delta}{2\sqrt{\delta^2-\delta+1}}\frac{1}{\cos\gamma}\begin{cases} \gamma = \frac{1}{6}\arccos\left(-1+\frac{27\delta^2(1-\delta)^2}{2(\delta^2-\delta+1)^3}\sin^2(3\theta_{\sigma})\right) & \text{for } \theta_{\sigma} \leq 0 \\ \gamma = \frac{\pi}{3}-\frac{1}{6}\arccos\left(-1+\frac{27\delta^2(1-\delta)^2}{2(\delta^2-\delta+1)^3}\sin^2(3\theta_{\sigma})\right) & \text{for } \theta_{\sigma} > 0 \end{cases} \tag{37}$$

In Eqs. (35)–(37), the aspect ratio δ can be determined from Eq. (28) and the Lode angle θ_{σ} from Eq. (18).

Table 3
True triaxial test data [8, 9,19] and predicted σ_1 for KTB Amphibolite.

σ_1 (MPa)	σ_2 (MPa)	σ_3 (MPa)	σ_1 (Z-Z)	σ_1 (S-D)	σ_1 (E-D)	σ_1 (H-D)
165	0	0				
346	79	0	364	231	207	235
291	149	0	458	286	283	286
347	197	0	500	518	360	365
267	229	0	256	415	410	416
410	30	30	455	455	455	455
479	60	30	510	529	506	529
599	100	30	572	576	535	580
652	200	30	688	602	559	611
571	249	30	730	592	557	600
637	298	30	763	572	561	579
702	60	60	638	638	638	638
750	88	60	682	684	687	700
766	103	60	702	724	703	725
745	155	60	767	781	740	784
816	199	60	815	807	759	813
*888	249	60	862	820	771	830
*828	299	60	902	822	776	833
*887	347	60	934	815	774	826
*954	399	60	962	800	767	810
*815	449	60	981	750	754	786
868	100	100	834	834	834	834
*959	164	100	915	941	919	942
*1001	199	100	953	978	945	981
*945	248	100	1004	1014	970	1020
*892	269	100	1022	1025	979	1033
*1048	300	100	1048	1038	988	1047
1058	349	100	1086	1050	999	1061
1155	442	100	1145	1050	1005	1064
1118	597	100	1201	1065	1052	1069
*1147	150	150	1041	1041	1041	1041
*1065	198	150	1097	1120	1109	1121
1112	199	150	1098	1121	1110	1122
1176	249	150	1150	1180	1154	1183
1431	298	150	1197	1222	1185	1227
1326	348	150	1239	1252	1208	1260
1169	399	150	1278	1274	1224	1284
1284	448	150	1312	1286	1235	1298
1265	498	150	1342	1292	1242	1306
1262	642	150	1407	1279	1239	1294
Prediction error (%)			9.98	8.99	9.84	8.18

Using one of these Lode dependences, Eqs. (35)–(37), to replace the Lode dependence of the generalized 3D Zhang–Zhu criterion, the modified criteria can be expressed as

$$\sqrt{J_2} = L(\theta_\sigma)_{X-D} \sqrt{J_{\max}} \quad (38)$$

where X is E , H , or S , respectively for the elliptical dependence, the hyperbolic dependence or the spatial mobilized plane dependence.

When $a=0.5$, the explicit expressions [Eqs. (19)–(33)] in the previous subsection can be used to calculate the aspect ratio and Eq. (38) can then be directly utilized to determine the strength with given σ_2 and σ_3 . In the case when $a \neq 0.5$ (corresponding to the more general condition for rock masses), however, an explicit expression for the modified criteria cannot be derived and thus a numerical solution strategy is proposed as below:

- Assume an initial σ'_1 with known σ_2 and σ_3 , say $\sigma'_1 = \sigma_2$;
- Calculate I_1 with Eq. (10);
- Calculate $J_{\max}$ and $J_{\min}$ from Eq. (23) with known m_b , s , a and using $\theta_\sigma = \pi/6$ for $J_{\max}$ and $\theta_\sigma = -\pi/6$ for $J_{\min}$;
- Calculate aspect ratio δ with Eq. (28);
- Calculate σ_1 with Eq. (38) using the Lode dependence from Eqs. (35–37);
- Check the error defined below

$$|\sigma_1 - \sigma'_1| < \varepsilon \quad (39)$$

when ε is the prescribed convergence limit (e.g., 0.001 MPa);

- If Eq. (39) is satisfied, the obtained σ_1 is the right σ_1 at given σ_2 and σ_3 from the modified criteria. If Eq. (39) is not satisfied, the obtained σ_1 is used as the σ'_1 and the previous steps are repeated until Eq. (39) is satisfied.

Table 4

True triaxial test data [9,31] and predicted σ_1 for Westerly granite.

σ_1 (MPa)	σ_2 (MPa)	σ_3 (MPa)	σ_1 (Z-Z)	σ_1 (S-D)	σ_1 (E-D)	σ_1 (H-D)
201	0	0				
306	40	0	340	293	246	296
301	60	0	390	296	251	299
317	80	0	433	291	253	295
304	100	0	471	282	252	285
231	2	2	235	235	235	235
300	18	2	295	303	269	304
328	40	2	360	323	284	336
359	60	2	408	341	291	345
353	80	2	450	340	295	345
355	100	2	487	335	295	339
430	20	20	446	446	446	446
*529	40	20	494	513	489	514
*602	60	20	536	552	510	554
*554	62	20	540	555	511	557
*553	61	20	538	553	511	555
*532	79	20	572	575	523	579
*575	100	20	609	592	534	597
*567	114	20	632	599	539	605
601	150	20	685	608	548	616
638	202	20	751	604	552	612
*605	38	38	593	593	593	593
*620	38	38	593	593	593	593
700	57	38	631	650	635	650
733	78	38	669	693	661	695
720	103	38	711	729	682	732
723	119	38	736	746	692	750
731	157	38	790	772	709	780
781	198	38	842	786	721	795
747	60	60	740	740	740	740
811	90	60	791	815	797	816
821	114	60	829	857	824	859
860	180	60	919	927	868	932
861	249	60	999	959	892	969
889	77	77	840	840	840	840
954	102	77	880	900	889	901
992	142	77	938	968	935	971
998	214	77	1029	1041	982	1047
1005	310	77	1129	1081	1013	1093
1012	100	100	962	962	962	962
1103	165	100	1054	1086	1058	1088
1147	167	100	1057	1089	1060	1091
1155	216	100	1117	1145	1097	1150
1195	259	100	1165	1180	1121	1187
1129	312	100	1218	1208	1142	1218
Prediction error (%)			9.99	5.35	7.10	5.66

The flow chart for this numerical solution strategy is shown in Fig. 1.

5. Validation of the modified criteria

In this section, the smoothness, convexity and prediction accuracy of the three modified criteria are evaluated by applying them to analyze both intact rocks and jointed rock masses. For comparison, the original generalized 3D Zhang–Zhu criterion is also applied in the analysis.

5.1. Smoothness and convexity

To validate the smoothness and convexity, the criteria are applied to analyze rocks with three different GSI values: GSI=100 for intact rock, and GSI=75 and 40 for rock masses. The input data for these three cases are listed in Table 2. The m_b , s , a are determined with Eq. (7). Figs. 2–4 show the π -plane plots of the generalized 3D Zhang–Zhu (Z-Z) criterion and the three modified criteria with respectively elliptical dependence (E-D), hyperbolic dependence (H-D) and spatial mobilized plane dependence (S-D), for these three rocks.

Table 5
True triaxial test data [9, 16] and predicted σ_1 for Dunham dolomite.

σ_1 (MPa)	σ_2 (MPa)	σ_3 (MPa)	σ_1 (Z-Z)	σ_1 (S-D)	σ_1 (E-D)	σ_1 (H-D)
257	0	0				
400	25	25	386	386	386	386
474	68	25	439	452	437	453
500	91	25	463	471	450	474
553	135	25	502	490	465	495
574	177	25	532	494	470	501
594	232	25	559	484	466	490
544	269	25	569	499	493	501
488	45	45	471	471	471	471
562	100	45	533	546	531	548
586	124	45	555	565	545	568
607	159	45	584	583	558	587
639	183	45	601	589	564	595
671	241	45	635	593	569	600
670	263	45	645	590	568	597
622	293	45	656	583	564	590
568	65	65	548	548	548	548
636	113	65	599	614	603	615
642	152	65	634	647	628	650
687	208	65	677	673	648	678
684	259	65	707	681	656	688
725	306	65	728	679	656	687
*700	390	65	748	692	684	695
624	85	85	618	618	618	618
682	126	85	661	675	668	676
718	150	85	683	699	686	701
743	230	85	745	747	723	752
*771	300	85	785	762	736	769
*818	371	85	813	757	736	766
*798	440	85	826	771	763	775
679	105	105	685	685	685	685
776	165	105	742	759	749	760
784	202	105	773	789	772	792
*804	265	105	818	821	798	827
*822	331	105	854	836	811	844
*839	351	105	864	837	812	845
*820	411	105	885	833	811	842
*863	266	105	818	822	798	827
724	125	125	747	747	747	747
823	186	125	804	821	812	822
*859	241	125	848	864	846	867
*862	288	125	880	887	864	892
*893	359	125	919	906	880	913
942	411	125	941	908	884	917
918	458	125	956	904	882	913
887	510	125	965	894	885	898
*892	254	145	901	919	903	921
*929	292	145	927	941	920	945
*924	319	145	944	953	930	958
922	342	145	958	961	937	967
1016	387	145	981	972	946	979
1003	404	145	989	974	949	982
953	451	145	1008	976	951	985
Prediction error (%)			4.17	4.79	6.44	4.48

Table 6True triaxial test data [7] and predicted σ_1 for Apache Leap tuff.

σ_1 (MPa)	σ_2 (MPa)	σ_3 (MPa)	σ_1 (Z-Z)	σ_1 (E-D)	σ_1 (H-D)	σ_1 (S-D)
147.3	0	0				
181.8	3.4	3.4	179	179	179	179
197	6.9	6.9	207	207	207	207
*240.4	10.3	10.3	232	232	232	232
*248.9	13.8	13.8	255	255	255	255
*250	13.8	13.8	255	255	255	255
*265.3	17.2	17.2	277	277	277	277
*242.3	20.7	20.7	297	297	297	297
308.4	24.1	24.1	317	317	317	317
329.8	27.6	27.6	336	336	336	336
356.7	31	31	353	353	353	353
360.2	34.5	34.5	371	371	371	371
396.5	37.9	37.9	387	387	387	387
419.5	41.4	41.4	404	404	404	404
456.1	44.8	44.8	420	420	420	420
434.1	48.3	48.3	435	435	435	435
464.9	55.2	55.2	465	465	465	465
190.1	18.4	0	190	192	173	193
*227.8	36.8	0	223	204	181	206
*241.1	55.2	0	250	205	184	208
286	73.5	0	272	199	184	203
325.9	110.3	0	305	214	212	214
341.3	92.3	0	290	191	184	193
333.7	79.9	0	278	197	183	200
272.6	61.5	0	258	203	184	207
276.6	50.7	0	243	205	184	208
*218.1	30.7	0	213	201	179	203
180.8	15.4	0	184	188	171	189
164.7	14.8	0	183	187	171	188
221.4	29.6	0	211	201	179	203
247.1	44.5	0	235	205	183	208
286.6	59.3	0	255	204	184	207
349.6	88.9	0	287	193	181	195
170.1	8.6	3.4	191	197	192	197
223.9	17.2	6.9	228	236	227	236
200.2	25.7	10.3	260	270	258	270
255.6	34.5	13.8	290	300	287	301
241.9	43.1	17.2	318	328	313	329
289.3	51.7	20.7	344	355	338	356
373.9	60.3	24.1	369	379	362	380
419.1	68.9	27.6	383	403	384	404
170.9	8.6	6.9	211	213	212	213
*242.3	12.9	10.3	237	240	239	240
201.7	17.2	13.8	261	265	264	265
*260.1	21.5	17.2	284	289	287	289
*244.8	25.9	20.7	306	311	309	311
328.8	30.2	24.1	326	332	330	332
295.9	34.5	27.6	346	352	350	352
340.3	38.8	31	365	371	367	371
Prediction error (%)			11.01	19.61	21.20	19.32

The smoothness and convexity of the three modified criteria can be obviously seen from the figures. The three modified criteria also predict the same strength as the Z-Z criterion at both triaxial compression and extension states. But the three modified criteria predict slightly higher strength at the zone near the triaxial compression state and slightly lower strength at the zone near the triaxial extension state than the Z-Z criterion. More detailed comparison between the three modified criteria and the Z-Z criterion is the applications presented in next subsection.

5.2. Prediction accuracy

5.2.1. Application to intact rocks

Zhang and Zhu [1] and Zhang [2] used the true triaxial compression test data of five rock types, KTB Amphibolite, Westerly granite, Dunham dolomite, Apache Leap tuff, Mizuho trachyte [7–9,16,19,31], to validate their proposed 3D Hoek–Brown criteria. The same test data as listed in Tables 3–7 are used here to validate the three modified criteria.

The values of parameter m_i for these five rock types are obtained by Zhang [2] based on best fitting with the generalized Zhang–Zhu (Z-Z) criterion. They are 31, 35, 10, 18 and 11 respectively for KTB Amphibolite, Westerly granite, Dunham dolomite, Apache Leap tuff and Mizuho trachyte. For intact rocks, $s=1$, and $a=0.5$. Using these m_i values, the σ_1 at the test σ_2 and σ_3 values are predicted using the three modified criteria, as listed in Tables 3–7, respectively for the five different rock types. For comparison, the predicted σ_1 from the

Table 7
True triaxial test data [9,16] and predicted σ_1 for Mizuho trachyte.

σ_1 (MPa)	σ_2 (MPa)	σ_3 (MPa)	σ_1 (Z-Z)	σ_1 (E-D)	σ_1 (H-D)	σ_1 (S-D)
100	0	0				
193	15	15	178	178	178	178
253	30	30	237	237	237	237
300	45	45	289	289	289	289
314	55	45	299	304	302	304
326	71	45	314	321	317	322
333	96	45	334	340	331	341
349	142	45	362	353	342	356
*361	214	45	380	365	360	367
*365	289	45	305	433	426	435
351	332	45	329	466	459	469
339	60	60	336	336	336	336
352	91	60	364	372	367	372
383	142	60	399	400	390	402
*396	191	60	421	406	396	410
*404	229	60	430	400	392	404
*400	271	60	428	383	379	386
383	331	60	371	340	340	340
365	75	75	379	379	379	379
400	114	75	412	421	416	421
*417	153	75	439	444	434	445
*438	229	75	472	455	445	459
439	300	75	477	435	430	438
424	343	75	458	410	408	411
451	391	75	379	366	366	367
*419	100	100	446	446	446	446
*460	137	100	477	485	481	486
489	186	100	509	515	506	517
494	274	100	545	530	520	534
522	382	100	541	496	493	498
513	411	100	522	476	476	478
Prediction error (%)			7.05	8.94	8.38	9.10

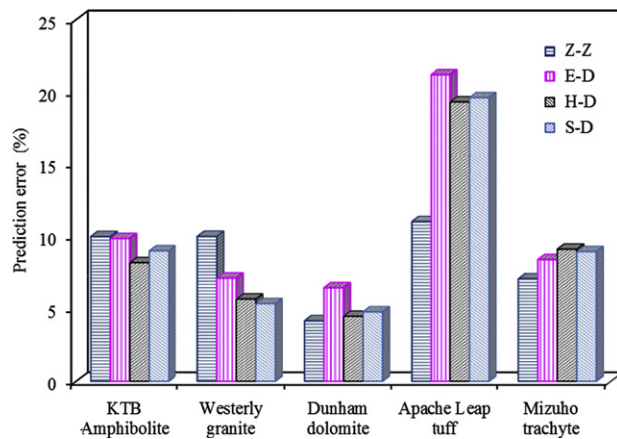


Fig. 5. Prediction errors for generalized Zhang–Zhu criterion (Z-Z) and three modified criteria with elliptical dependence (E-D), hyperbolic dependence (H-D), and spatial mobilized plane dependence (S-D) for five rock types.

Table 8
Values of σ_c , m_i and s used for predicting σ_1 of five intact rocks with generalized Zhang–Zhu criterion and three modified criteria.

Rock	σ_c (MPa)	Range of m_i^*	m_i based on best fitting				s
			Z-Z	S-D	E-D	H-D	
KTB amphibolite	165.0	26 ± 6	31	32	32	32	1
Westerly granite	201.0	32 ± 3	35	35	35	35	1
Dunham dolomite	257.0	9 ± 3	10	11	11	10	1
Apache leap tuff	147.3	13 ± 5	18	18	18	18	1
Mizuho trachyte	100.0	13 ± 4	11	12	12	12	1

* Based on Table 1.

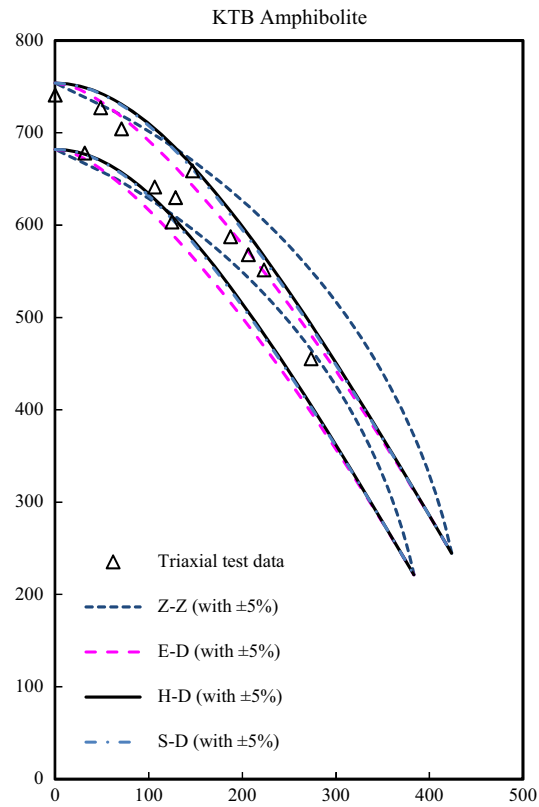


Fig. 6. Failure surface plots of generalized Zhang–Zhu (Z-Z) criterion and three modified criteria with elliptical dependence (E-D), hyperbolic dependence (H-D), and spatial mobilized plane dependence (S-D) at $I_1/3=439$ MPa (with $\pm 5\%$) for KTB Amphibolite.

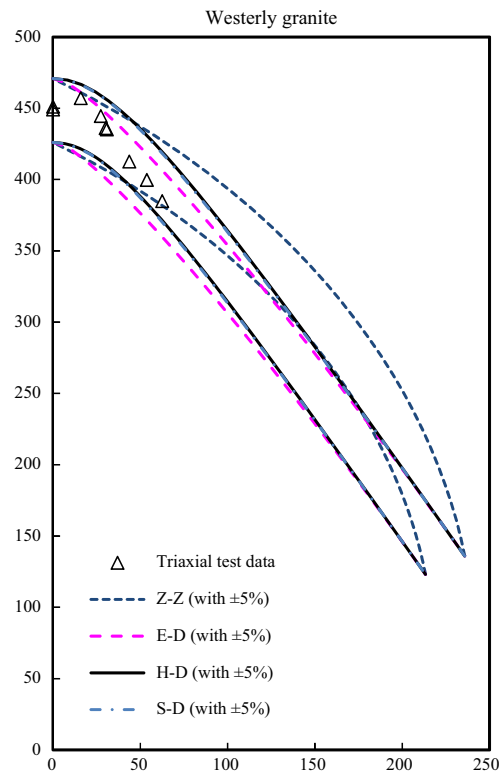


Fig. 7. Failure surface plots of generalized Zhang–Zhu (Z-Z) criterion and three modified criteria with elliptical dependence (E-D), hyperbolic dependence (H-D), and spatial mobilized plane dependence (S-D) at $I_1/3=220$ MPa (with $\pm 5\%$) for Westerly granite.

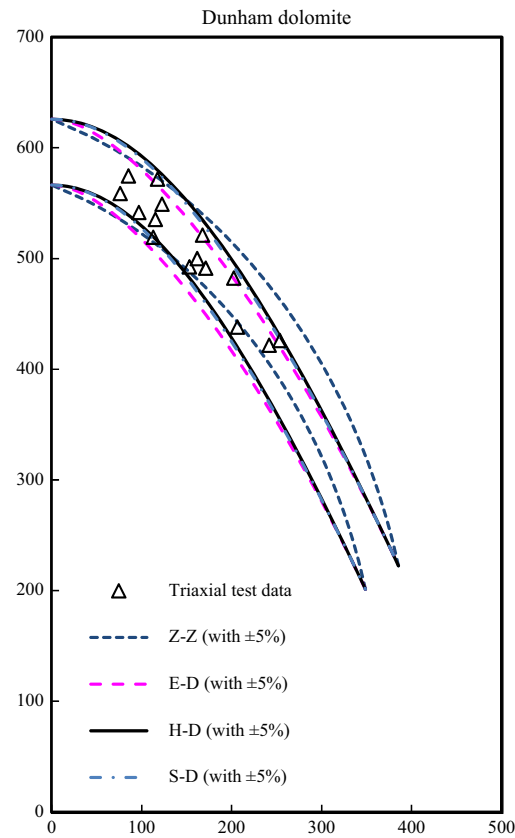


Fig. 8. Failure surface plots of generalized Zhang–Zhu (Z-Z) criterion and three modified criteria with elliptical dependence (E-D), hyperbolic dependence (H-D), and spatial mobilized plane dependence (S-D) at $I_1/3=425$ MPa (with $\pm 5\%$) for Dunham dolomite.

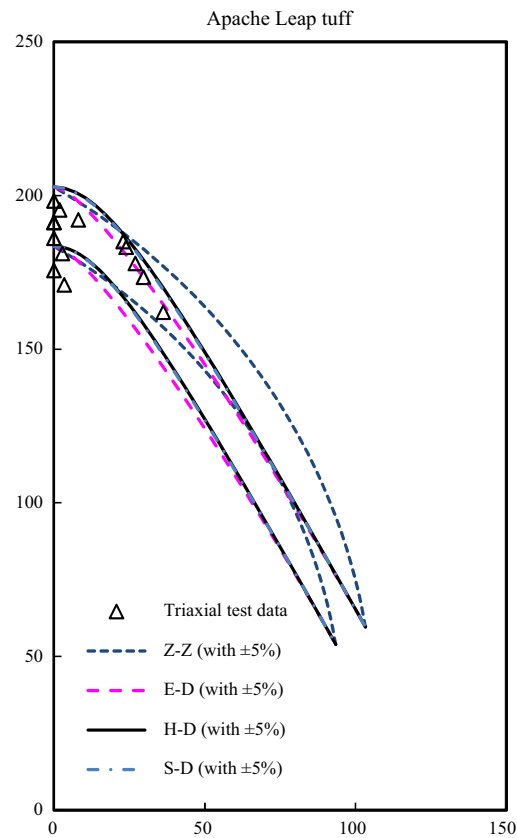


Fig. 9. Failure surface plots of generalized Zhang–Zhu (Z-Z) criterion and three modified criteria with elliptical dependence (E-D), hyperbolic dependence (H-D), and spatial mobilized plane dependence (S-D) at $I_1/3=92$ MPa (with $\pm 5\%$) for Apache leap tuff.

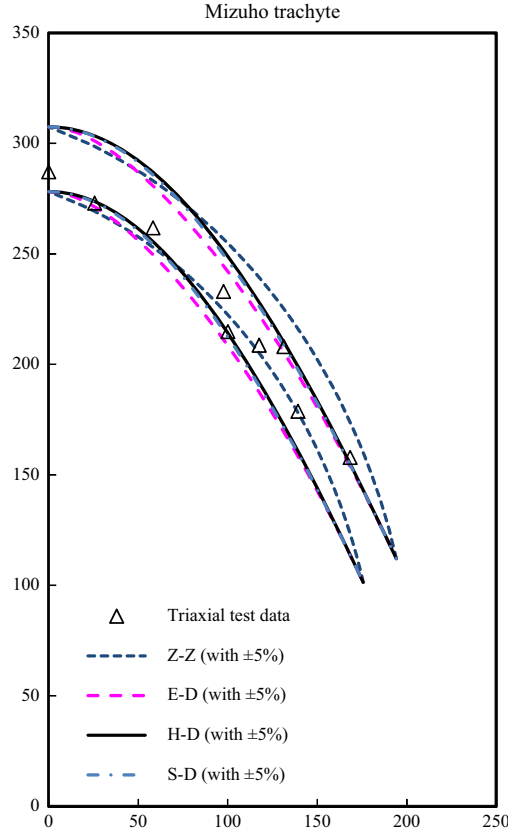


Fig. 10. Failure surface plots of generalized Zhang–Zhu (Z-Z) criterion and three modified criteria with elliptical dependence (E-D), hyperbolic dependence (H-D), and spatial mobilized plane dependence (S-D) at $I_1/3=227$ MPa (with $\pm 5\%$) for Mizuho trachyte.

Z-Z criterion are also listed in the tables. The prediction errors listed in the tables are determined by

$$S = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (\sigma_{1,\text{predicted}} - \sigma_{1,\text{test}})^2}}{(\sigma_{1,\text{test}})_{\text{average}}} \quad (40)$$

where n is the number of test data points. The results (see also Fig. 5) show that the three modified criteria have about the same prediction accuracy as the Z-Z criterion.

To further evaluate the accuracy of the three modified criteria, the method of least squares (LS) is used to find the best fitting (i.e., finding the value of m_i so that the error defined by Eq. (40) is the minimum) for each criterion. The obtained best fitting m_i values, including those for the Z-Z criterion, are listed in Table 8. It can be seen that best fitting m_i values are equal or very close to each other for all criteria. This further shows that the three modified criteria have about the same prediction accuracy as the Z-Z criterion.

To study the shape of the failure surface plots on the π -plane, several test data points with the same first stress invariant I_1 are required. Since the test data contains no points with exactly the same I_1 value, the points with I_1 within less than 10% discrepancy are selected. These test data points are labeled with '*' in Tables 3–7. Figs. 6–10 show the failure surface plots of the generalized 3D Zhang–Zhu (Z-Z) criterion and the three modified criteria with the elliptical dependence (E-D), the hyperbolic dependence (H-D), and spatial mobilized plane dependence (S-D) respectively for KTB Amphibolite, Westerly granite, Dunham dolomite, Apache Leap tuff and Mizuho trachyte. For each criterion, two failure surface plots respectively at $\pm 5\%$ of I_1 are also shown in the figure. The failure surfaces of the modified criteria are close to that of the Z-Z criterion, especially in the zone near the triaxial compression state. It can also be seen that the three modified criteria and the Z-Z criterion all predict the strength in a good accuracy, although the Z-Z criterion tends to over-predict the strength at the zone close to the triaxial extension state (when $\theta_\sigma < 0$).

5.2.2. Application to rock masses

The true triaxial test data of a jointed biotite granite at the Kurobe IV arch dam site in Japan [18,32] was studied by Zhang and Zhu and Zhang [1,2]. The test block was 2.8 m long in the major principal stress direction, 2.8 m high in the intermediate principal stress direction, and 1.4 m wide in the minor principal stress direction, giving it a total volume about 11 m^3 . The jointed biotite granite was described as 'somewhat decomposed' and the unconfined compressive strength of granite cores taken from the test location was 23.0 MPa. The test was conducted by increasing and cycling σ_1 with σ_2 and σ_3 held constant at 0.70 and 0.12 MPa, respectively. The failure stress was 13.0 MPa which is much lower than the unconfined compressive strength of the corresponding intact rock.

According to Table 1, $m_i=29\text{--}35$ for granite. For simplicity, an average value of $m_i=32$ is selected for the analysis. The GSI value of about 81 is selected based on the fitting of the measured σ_1 with the generalized 3D Zhang–Zhu criterion [2]. With the selected m_i and GSI, $m_b=16.115$, $s=0.118$, and $a=0.5005$ can be obtained from Eq. (7) by assuming that the tested rock mass is undisturbed (i.e., $D=0$).

With these obtained parameters, the predicted failure stress σ_1 with the E-D, H-D and S-D criteria are respectively 11.8, 13.4 and 13.4 MPa, which are only 9.2%, 3.1% and 3.1% different from the test data 13.0 MPa. So the strength prediction with the modified criteria for rock masses is quite reliable.

6. Summary and conclusions

This paper evaluated the generalized 3D Zhang–Zhu criterion and proposed modification to address its non-smoothness and non-convexity problem. Based on the study, the following conclusions can be drawn.

The reason for the non-smoothness and non-convexity of the generalized 3D Zhang–Zhu criterion is because its Lode dependence does not meet the requirements of Lode dependence regarding smoothness and convexity. This is a common problem for the strength criteria based on the concept of Mogi [16] criterion.

Three Lode dependences with characteristics of both smoothness and convexity, the elliptical dependence (E-D), the hyperbolic dependence (H-D) and the spatial mobilized plane dependence (S-D), were used to modify the generalized 3D Zhang–Zhu criterion. The modification completely solves the non-smoothness and non-convexity problem of the generalized 3D Zhang–Zhu criterion.

A numerical solution strategy was proposed for the case when $a \neq 0.5$ so the modified criteria can be applied to general rock masses.

The three modified criteria not only solve the non-smoothness and non-convexity problem, but keep the advantages of the generalized 3D Zhang–Zhu strength criterion and have about the same good accuracy for strength prediction.

Acknowledgments

The first author wishes to thank the China Scholarship Council (CSC) for supporting him to do research on rock mechanics in the Department of Civil Engineering and Engineering Mechanics at the University of Arizona as a visiting scholar. The research is supported by State Key Laboratory for GeoMechanics and Deep Underground Engineering, China University of Mining & Technology, with Open Research Fund (SKLGDUEK1011).

References

- [1] Zhang L, Zhu H. Three-dimensional Hoek–Brown strength criterion for rocks. *J Geotech Geoenviron Eng ASCE* 2007;133(9):1128–35.
- [2] Zhang L. A generalized three-dimensional Hoek–Brown strength criterion. *Rock Mech Rock Eng* 2008;41:893–915.
- [3] Bieniawski ZT. Estimating the strength of rock materials. *J S Afr Inst Min Metall* 1974;74:312–20.
- [4] Hoek E, Brown ET. Empirical strength criterion for rock masses. *J Geotech Eng ASCE* 1980;106:1013–35.
- [5] Johnston IW. Strength of intact geomechanical materials. *J Geotech Eng* 1985;111:730–49.
- [6] Lade PV. Rock strength criteria: the theories and the evidence. In: Hudson JA, editor. *Comprehensive Rock Engineering*. Oxford: Pergamon Press; 1993. pp. 255–284.
- [7] Wang R, Kemeny JM. A new empirical criterion for rock under polyaxial compressive stresses. In: Daemen JJK, Schultz RA, editors. *Rock mechanics*. Rotterdam: Balkema; 1995. pp. 453–458.
- [8] Chang C, Haimson B. True triaxial strength and deformability of the German Continental deep drilling program (KTB) deep hole amphibolite. *J Geophys Res* 2000;105:8999–9013.
- [9] Al-Ajmi AM, Zimmerman RW. Relation between the Mogi and the Coulomb failure criteria. *Int J Rock Mech Min Sci* 2005;42(3):431–9.
- [10] Zhang L. *Engineering Properties of Rocks*. Amsterdam, Netherlands: Elsevier; 2005.
- [11] Hoek E, Brown ET. The Hoek–Brown criterion—a 1988 update. In: *Proceedings of the 15th Canada rock mechanics symposium*. University of Toronto, Canada; 1988. p. 31–8.
- [12] Hoek E. Strength of rock and rock masses. *ISRM News J* 1994;2(2):4–16.
- [13] Hoek E, Kaiser PK, Bawden WF. Support of Underground Excavations in Hard Rock. Rotterdam: Balkema; 1995.
- [14] Hoek E, Carranza-Torres C, Corkum B. Hoek–Brown failure criterion—2002 edition. In: *Proceedings of the 5th the North American rock mechanics society symposium*. Toronto; 2002. p. 267–73.
- [15] Hoek E. Rock engineering—course notes by Evert Hoek, editor. <<http://www.rocscience.com/hoek/PracticalRockEngineering.asp>>; 2006.
- [16] Mogi K. Fracture and flow of rocks under high triaxial compression. *J Geophys Res* 1971;76(5):1255–69.
- [17] Pan X, Hudson JA. A simplified three dimensional Hoek–Brown yield criterion. In: Romana M, editor. *Rock Mechanics and Power Plants*. Rotterdam: Balkema; 1988. pp. 95–103.
- [18] Brown ET. The nature and fundamentals of rock engineering. In: Hudson JA, editor. *Hudson Comprehensive Rock Engineering*. Oxford: Pergamon; 1993. pp. 1–23.
- [19] Colmenares LB, Zoback MD. A statistical evaluation of intact rock failure criteria constrained by polyaxial test data for five different rocks. *Int J Rock Mech Min Sci* 2002;39(6):695–729.
- [20] Priest SD. Determination of shear strength and three-dimensional yield strength for the Hoek–Brown yield criterion. *Rock Mech Rock Eng* 2005;38(4):299–327.
- [21] Melkounian N, Priest SD, Hunt SP. Further development of the three-dimensional Hoek–Brown yield criterion. *Rock Mech Rock Eng* 2009;42(6):835–47.
- [22] Bardet JP. Lode dependence for isotropic pressure-sensitive elastoplastic materials. *Trans ASME* 1990;57(9):498–506.
- [23] Lin F, Bazant ZP. Convexity of smooth yield surface of frictional material. *J Eng Mech ASCE* 1986;112:1259–62.
- [24] Jiang J, Pietruszczak S. Convexity of yield loci for pressure sensitive materials. *Comput Geotech* 1988;5(1):51–63.
- [25] Drucker DC. A more fundamental approach to plastic-stress strain relations. In: *Proceedings of the 1st US national congress applied mechanics*. New York; 1951. p. 487–91.
- [26] You M. *Mechanical properties of rocks*. Beijing: Geology Press; 2007.
- [27] Benz T, Schwab R. A quantitative comparison of six rock failure criteria. *Int J Rock Mech Min Sci* 2008;45(7):1176–86.
- [28] William KJ, Warnke EP. Constitutive model for the triaxial behavior of concrete. In: *International association for bridge and structural engineering*. Bergamo; 1975.
- [29] Yu M. *New system of strength theory*. Xian: Xian Jiaotong University; 1990.
- [30] Matsuoaka H, Nakai T. Stress-deformation and strength characteristics of soil under three different principal stress. In: *Proceedings of Japan society of civil engineers*. Japan; 1974. p. 59–70.
- [31] Haimson B, Chang C. A new true triaxial cell for testing mechanical properties of rock, and its use to determine rock strength and deformability of Westerly granite. *Int J Rock Mech Min Sci* 2000;37(1):285–96.
- [32] Müller-Salzburg L, Ge X. Untersuchungen Zum Mechanischen Verhalten Geklüfteten Gebirges unter Wechsellasten. In: *Proceedings of the 5th congress international society of rock mechanics*. Rotterdam; 1983. p. 43–49.
- [33] Hoek E, Brown ET. Practical estimates of rock mass strength. *Int J Rock Mech Min Sci Geomech Abstr* 1997;34:1165–86.
- [34] Marinos P, Hoek E. Estimating the geotechnical properties of heterogeneous rock masses such as flysch. *Bull Eng Geol Environ* 2001;60:85–92.